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~~Turner~~ Problem - Picture ~~Intermittent~~ and Sound
Separated from Picture

When fine tuning is adjusted for the best picture, if the sound is over and vice versa, the problem can be caused by poor alignment or problems in the Afc System or the I.F Stages.

Whenever tuner response is poor with Afc network and I.F stages cleared. The problem cannot be corrected by alignment.

The defects are

Open neutralizing capacitors,

Open bypass capacitors,

Loose or poorly grounded tuner Shields.

Any of these ~~can~~ must likely defects can cause poor tuner operation, but will not show up as incorrect d.c. voltage on the transition elements.

1
2
3

C 203 Shorted, no Sound

1. Establish that 1-C 225 is operating correctly (by turning up the volume potentiometer and touching Pin 8 of 1-C 225).

2. Voltage measurements on 1-C 206 Pins 2, 5, and 11 will show that the automatic circuit, the volume control and the operating voltage are all okay.

3. No voltage will be present at Pin 13.

(b) C 218 Shorted to earth, no Sound.

1. When Pin 8 of 1-C 225 is touched, there will be no hum in the loudspeaker.

2. Voltage measurements on 1-C 225 will show that 0V is present in Pin 7 instead of 3V.

(c) C 202 Shorted, no AF only hum.

1. The fault is likely to be located in the audio IF stage.

2. Voltage measurements on 1-C 206 will show that the voltage at Pin 14 is too high (approx 3V).

3. A short circuit will be indicated if the ceramic filter is checked using the ohmmeter.

(d) R 231 has high impedance. No Sound.

(1) No voltage is present on 1-C 225

2. The operating and supply voltage should always be measured first. It will show that 115 (17.2V) is present.

(3) The voltage will be missing following R 231 at Pin 2 of 1-C 225.

4. Since R 231 does not heat up, a short circuit in condenser C 229 or C 228 can be excluded.

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(E) R217 has impedance. No Sound.

① There will be no voltage at Pin 11 of IC206.

② Since R217 does not heat up. A Short ~~Circuit~~ Circuit in C015, C016 can thus be excluded.

③ Touch Pin 8 of IC205 via a conductive object, the result will be loud humming in the loudspeaker.

④ A fault in and around the output stage IC can thus be generally be excluded.

⑤ Test R217 with the ohmmeter, an interrupt in R217 will be revealed.

(F) Moving cord in the loudspeaker interrupted. No Sound.

① Operating and supply voltage and all voltages at ICs 206 and 255 are correct.

② The fault must therefore be located in the output stage IC, at its external connection or in the loudspeaker.

The electronics of a tape recorder are, A.F. amplifier, tone compensation circuits and biasing when recording. The ^{Major} difficulty encountered is the mechanical adjustments (alignment) which can make or make both recording and reproduction. Magnetisation of tape heads can cause noisy operation, and to ~~ensure~~ must be exercised in removing these deposits, never use Swivel cleaner but use simple wetting latex spirit. Spray a little on the heads before moistening the cloth. Use some products to clear oxide deposits from tape guides, capstan spindle, pinch roller and pressure pads.

If the pressure pads get solid. Some flutter and squeaking may occur. Soak them thoroughly with wetting latex spirit and roughen their surface with blade or pen knife. Don't use glass or emery paper.

The method of tape head alignment varies with the type of deck employed, but if they are badly out of alignment on four track recorders, reproduction of the tracks will not be equal. If tracks 1 and 4 are weak compared with 2 and 3 the head is too high, and conversely, if 2 and 3 are weak compared with 1 and 4 the head is low.

→ Always be careful when NOTE removing the cover plate to gain access to the interior. Some screws don't have to be removed, and it can be embarrassing to find a lot of odd parts floating loose.

Good Sound in Radio Side But Low Sound when Playing Cassette

- a Tear the audio cassette in another tape recorder.
 - b Check if the ~~tape~~ ^{cassette} inside the ~~tape~~ ^{cassette} is not well positioned on the play head.
 - c Clean the play head
 - d Check if the play head is worn out
 - e Align the play head for better sound.
- 2 Good Play But no Recording.
 - a Check the cassette, if the record safety tab has been removed.
 - b Check if there is no proper contact to the circuit recording unit.
 - c Wash the circuit switch with switch cleaner.
 - d Check the record play.

3 Low Sound on Play

- a Check for low voltage
- b Check the play mechanism
- c Test the audio cassette ^{with} another tape recorder.
- d Replace the drive belt.
- e The play motor is faulty.

73 AUDIO AMPLIFIER OR PUBLIC ADDRESS SYSTEM (SLAVE)

NO SOUND / INTERMISSION Lamp ON.

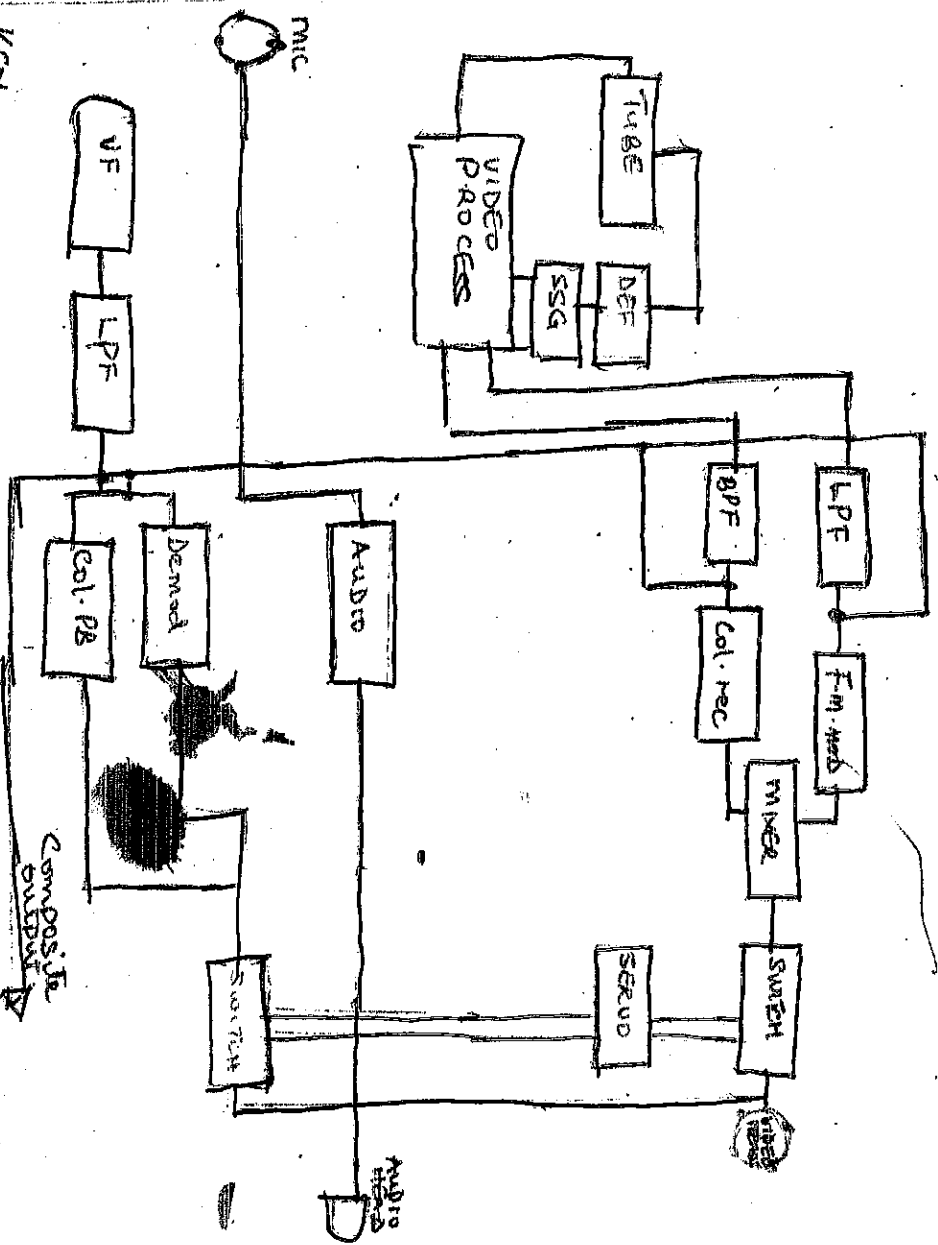
1. Speaker fuse blown
2. Function relay switch refuse to close.
3. Microphone or Signal Source open.
4. Open or overheating of a passive component.
5. Blown Speakers or loudspeaker connect.
6. Open circuit in volume control
7. ~~Open~~ Shorted Power transistor or I.C.

DISTORTED SOUND

1. Check the I.C. or power transistor for overheating
2. wrong fixed & component polarity
3. Shorted electrolytic capacitor
4. loudspeaker or pad microphone cable Shorted.
5. Wrong connections of the audio amplifier.
6. Lack of proper earthing.
7. Shorted loudspeaker.
8. very low voltage.

7. In correct colour
- (a) (all colour present)
 - (b) Check phase shift network between 3.58 MHz.
 - (c) Check oscillator and demodulation.

VIDEO CAMERA



KEY:

SSG = Sync Signal generator, LPF = low pass filter

BPF = Band pass filter, SSG = Sync generator

Block Diagram of A CAMERA AND RECORDER SECTIONS OF A VIDEO CAMERA.

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BLACK & WHITE AND COLOUR TELEVISION: ACCESSION FROM SLIDESHOW

TROUBLE (FAULT)

REMEDIES:

1. Receiver dead, no sound, no picture & filament.

- ① Check on 10T switch.
2. Check relay switch
3. Check line fuse or circuit breaker.
4. If series filament circuit,
5. Defective tube
6. Check a.c power outlet.

2. Receiver dead, filament on.

1. Check low-voltage fuse (H.T. fuse)
2. Check low-voltage rectifier
3. Low volt. Surge resistor may be open.

3. Dark horizontal bar across screen.

1. open filter capacitor in low power supply.
2. check rectifier diodes
3. Check for horizontal short in video, i.e. tuner or video section.

4. No brightness on screen, sound okay.

1. listen for high volt. cracking sound a minute, after set is turned on, if cracking is heard, high volt. is present.
2. If no cracking sound is heard check high voltage rectifier, check horizontal oscillator and amplifier.
3. check high voltage fuse.

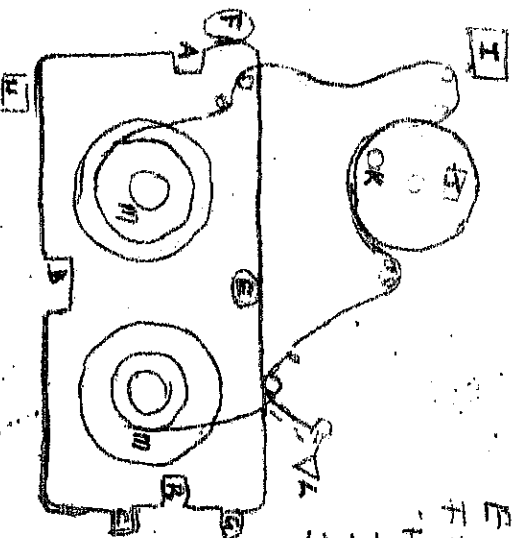
5. Picture size varies as brightness changes.

1. check HV regulator control
2. Check volts in HV regulator
3. Use HV probe and voltage to check for proper HV regulator setting.

6. Incorrect colour on any colour missing.

1. Check demodulation and check the voltage at chrominance (R&B).
2. Check transistor or i.c and their leads components.
3. Determine which colour is missing and investigate specific section, especially helpful in colour beam pattern is especially helpful.
4. A colour beam pattern is especially helpful.
5. Check for open filament in one gun or colour tube.

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TYPICAL POSITIONING OF DECK SENSORS IN A FRONT LOADING VHS, VCR AND VIDEO CAMERA. (DESIGN VARIES BETWEEN MAKES AND MODELS).

In a front loading VCR or a Cam Corder/Video Camera recorder, the first operation is insertion of a cassette, which closes one or two cassette - in switches in the cradle, A and B in figure above. So long as power is available above the cassette in and down when the cassette is fully down the Hi motor is switched off by a limit - switch C generally mounted on the cradle itself or sometimes by the action of the cassette body interrupting a light path between the cassette lamp E and the tape end sensors F and G.

With the cassette in, the machine is ready to go.

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VIDEO CAMERA

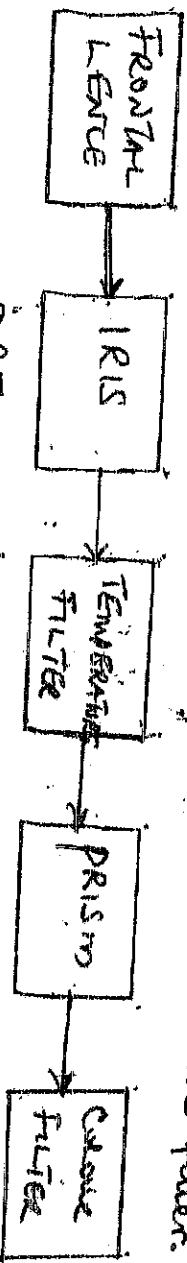
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The video camera consists of three major parts, all housed in same compartment. These parts include:

- (a) The optical lens system
- (b) The tube (the conversion unit)
- (c) The amplifier (the recorder).

OPTICAL LENS UNIT.

The optical system consists of the frontal lens, iris, temperature filter, prism and colour selective filter.



OPTICAL LENS STAGES.

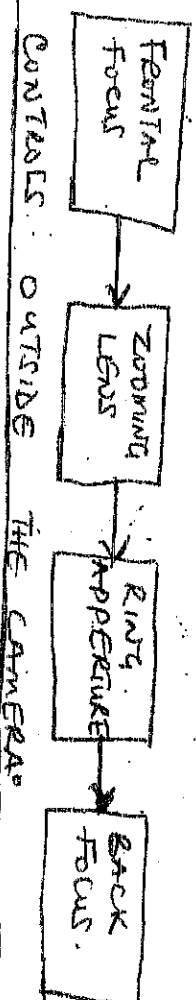
The function of each stage of optical lens are as follows:
FRONTAL LENS is used to focus the desired scene on the light sensitive device (tube).

IRIS: This is used to regulate the amount of light to the tube.

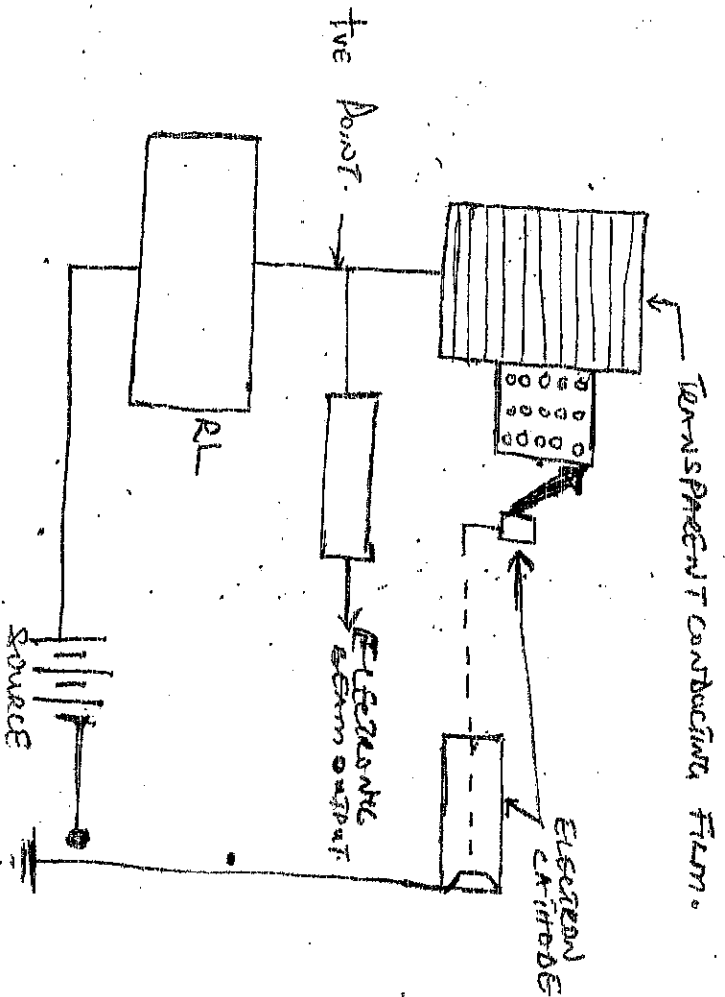
TEMPERATURE FILTER: This is used to translate the mood of the day (light) into temperature and also corrects various light conditions.

PRISM: This serves in reflecting different wave length of light through different angles (R.O.T.A.S.I.V), Red, orange, yellow, green, blue, indigo, violet.
COLOUR SELECTIVE FILTER is used to filter individual signal wavelength.

Attached to the optical lens unit are the following controls at the lens barrel of the camera.



TUBE UNIT.



TUBE UNIT OF A TELEVISION CAMERA

The visible scene to be converted (recorded) is focused by the optical lens system of the TV camera onto the light sensitive surface which absorbs light energy in accordance to the instantaneous scene being focused by the TV camera.

To televise a scene in colour, however three camera tube are required. All housed in the same compartment, each tube record the amount of each primary colour in the scene, in other camera additional tube is added, the fourth is used to pick-up tube with all colour filtering to record the brightness or luminance signal only as in the monochrome systems.

This information on the target in TV camera tube is extracted by a process known as scanning. This means the TV systems are based on translating the scene to be televised into a series of small signal points turned up to picture elements.

The colour TV camera has two side effects:

1. Registration
2. Colour balancing

Registration: Each tube is scanned by its internal electron beam and it is important that at any given instant the beams in each tube are looking at the same picture elements. The scanning beam in the pick-up tube should be line up so as to coincide or register on the image focused on the face plate.

Any loss of registration will mean that the three colour image will not coincide in time the camera output or in the position on the screen. This problem arises in multi tube camera, because colour separation is carried out before the scanning process.

2. COLOUR BALANCING ~~OR WHITE BALANCE~~

Colour balancing process is becoming of the ideal ratio of primary colour (RGB) to acquire with exact colour of the domestic receiver.

It should be noted that our video distribution normally is carried on a coaxial cable of 75 Ω nominal impedance.

The arrangement of the colour bars in colour television or pattern generator are:

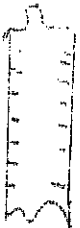
White, Yellow, Cyan, Green, Magenta, Red, Blue, Black.

At Completion of the tape loading, the following actions should occur almost simultaneously, Loading motor stop reel motor start driving the take up Spool, and tape drive begins by either the pinch roller closing to the capstan or the capstan motor turning on. All these actions are triggered by the loading end sensor. If the cassette could not (settle down) be loaded, check A and B switches, check for power to the System Control (Syscon) I.C. and check the loading motor. Whatever the form of limit switch, its failure will generally result in the cassette being ejected a few seconds after its insertion.

~~Known~~ Problems

- an incorrect aligned mechanism or Switch can generally cause in VCR or Camcorder/VCR Camera.
1. Failure of the deck to operate in one or modes.
 2. Deck's Shutdown after an attempt at a mechanical function.
 3. Oscillation of the mechanism.
 4. A growling or clicking noise from an over loaded and tooth jamming pinion, worn or bent.

TAPE DAMAGED PATTERN



- Causes
- ① Bidding against the lower drum
 - ② Spool carrier or a guide, and usually
 - ③ upset the signal transfer of contact pulses.

Remedy

- ① check for spools loading
- ② guide alignment
- ③ gently pinch roller or
- ④ excessive force up torque.

[Signature]

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Causes ① Sharp abrasive particles on some
stationary.

② tape-path member.

Remedy

① check a fixed guide

② The lower drum

③ FE head or

④ And in head stack.

Causes ① The offender is on a rotating component

Remedy ① check the capstan

② A guide sleeve or

③ The pinch roller in ascending order
of above or check spacing.

Causes ① Capstan

② Pinch roller.

③ If jitter in one setting of the tape

Suspect a crumpled tape-up torque.

Remedy ① Check the loading arms/guide

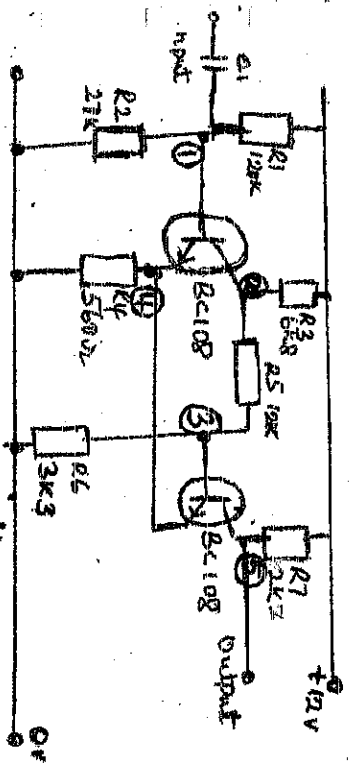
② check reel reverse operation

③ Unloading tape-up torque.

Causes: incorrect height setting of guide rollers or
guide pole.

Remedy: Re-alignment of the above components.

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TP	1	2	3	4	5
mV	1.5	0.95	0.2	0.9	12

Schmitt trigger circuit

The circuit is designed to produce square waves from a 1 kHz 1V peak-peak sine wave input. The potential divider R_1, R_2 sets up a voltage on the base of T_1 that just exceeds the threshold of the circuit. Thus in the stable state T_1 conducts and T_2 is held off. Only when the sine wave input actually falls below about -0.2V will the circuit change state with T_1 off and T_2 on. The circuit continuously switches back and forth as the sine wave input changes, but since the switching is so rapid a square wave output is produced. The amplitude is about 5V.

Questions

1. What effect on d.c. voltage and output waveform would result if R_1 went open circuit?
2. In each of the following faults the circuit fails to produce an output signal when the faulty component is applied. State, with reasons, the faulty component.

P.T.O

[Signature]

Test Point	1	2	3	4	5
Fault A	1.5	8.1	1.8	1.15	6.4
B	1.45	0.95	0.4	0.85	12
C	2	8.15	1.7	1.7	12
D	1.4	0.85	0.8	0.8	12
E	1.6	1.6	0.2	0.9	12
F	1.45	0.95	0	0.85	12

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